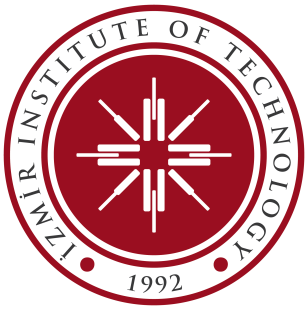


İzmir Institute of Technology

Faculty of Engineering

Department of Electrical and Electronics Engineering

EE 315 Electronics Laboratory Prelab Sheet

Experiment 2

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1 LTspice Simulation

1.1 Full-Wave Bridge Rectifier

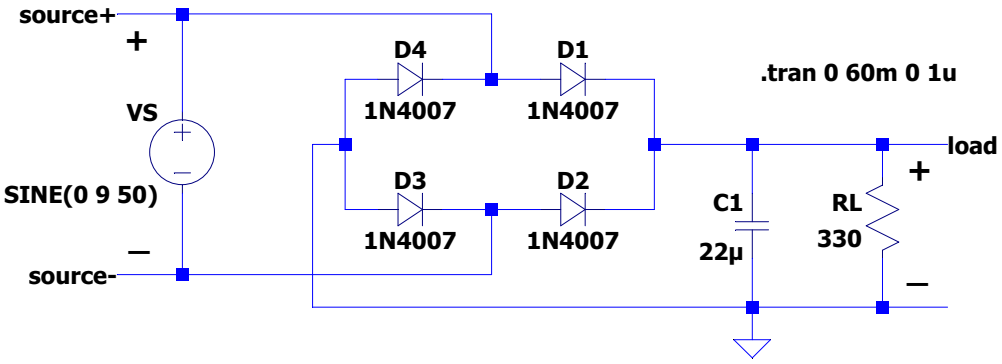


Figure 1: Schematic of full-wave bridge rectifier circuit

R_L	$V_{L,\text{peak}}$	f_L	**Number of conducting diodes**
330 Ω	7.59 V	100 Hz	2

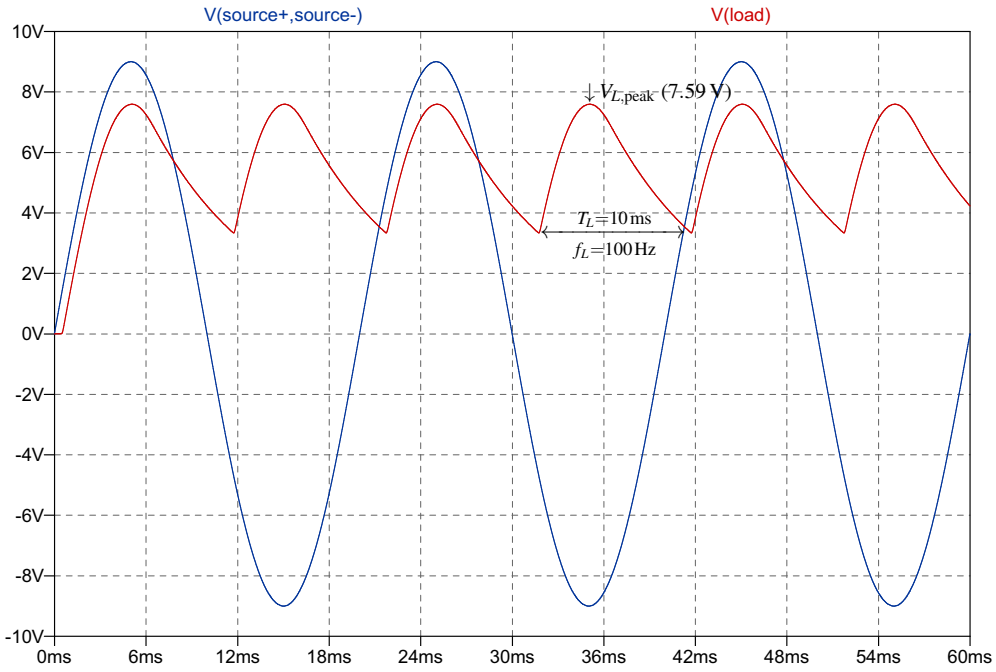


Figure 2: Input and output voltage waveforms of the full-wave bridge rectifier circuit ($R_L = 330\,\Omega$)

1.2 Zener Diode Regulator

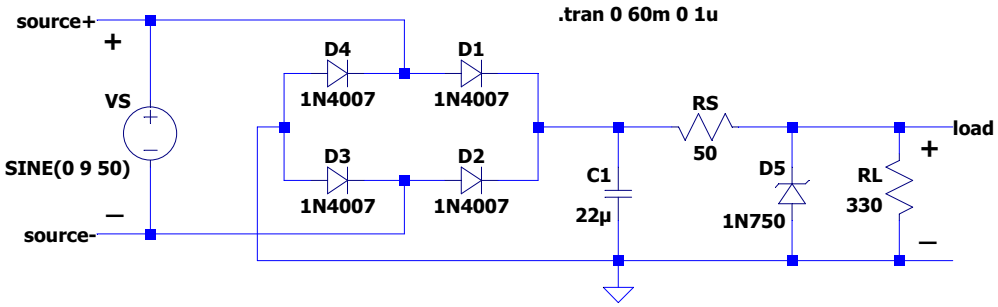


Figure 3: Schematic of Zener diode regulator circuit

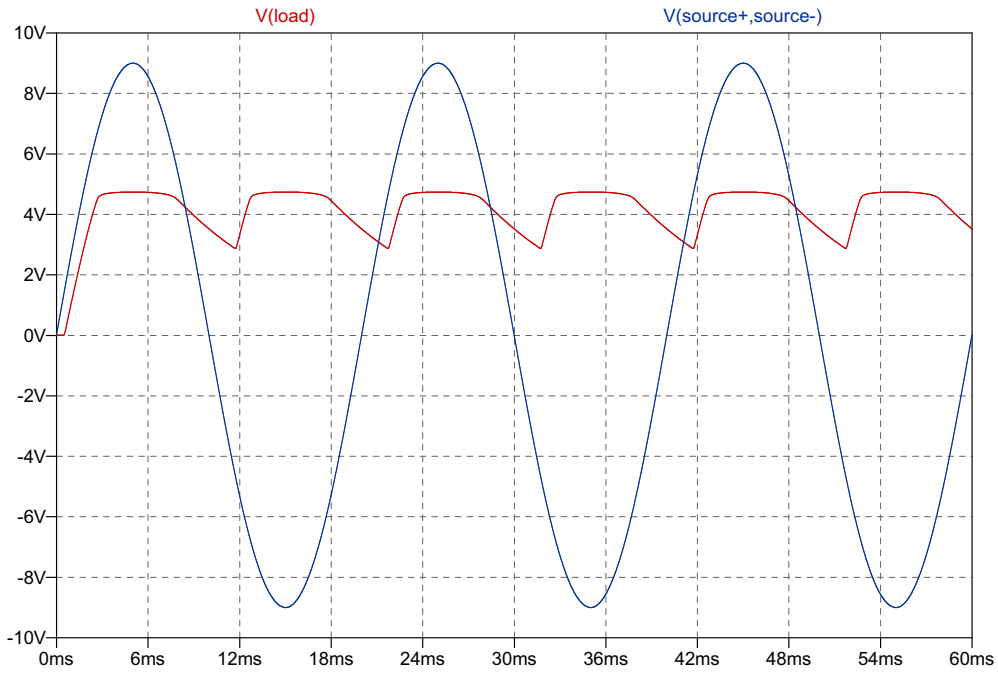


Figure 4: Input and output voltage waveforms of the Zener diode regulator circuit ($R_L = 330\,\Omega$)

Breakdown Voltage (V_Z)	$V_{o,DC}$	I_Z
4.7 V	4.13 V	-13.15 mA